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Article in SPE Reservoir Evaluation & Engineering · August 2003

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Modeling of Multiple Echo-Time NMR Measurements for Complex Pore Geometries and Multiphase Saturations

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Summary

We develop a numerical algorithm to simulate nuclear magnetic resonance (NMR) measurements in the presence of constant magnetic field gradients. The algorithm is based on Monte Carlo conditional random walks in restricted and unrestricted space. Simulations can be performed of 3D porous media that include both arbitrary bimodal pore distributions and multiphase fluid saturations. The ability to account for the presence of a constant external magnetic field gradient allows us to replicate actual well-logging conditions that include the effect of Carr-Purcell-Meiboom-Gill (CPMG) pulse sequences at a microscopic level. This is accomplished by simulating ideal pulse-acquisition techniques that include multiple interecho times (TE) similar to those currently used by the well-logging industry. Benchmark examples are presented to validate the accuracy and internal consistency of our algorithm against previously published results for the case of a null magnetic field gradient. Validation examples also are presented against actual NMR measurements performed on core samples of carbonate rock formations.

Interpretation work is focused on the petrophysical assessment of both partial oil/water saturations and pore structures exhibiting diffusive coupling. Simulation examples are designed to quantify whether the inclusion of diffusion under a magnetic field gradient can improve the interpretation of multiphase fluid saturations when diffusion coupling is significant. The simulation algorithm sheds light on new NMR data-acquisition strategies that could be used to improve the detection and quantification of fluid types, complex fluid saturations, and complex pore geometries.

Introduction

The presence of diffusion pore coupling in carbonate rocks (mainly grainstones) challenges conventional NMR interpretation techniques. Although pore-coupling phenomena are commonplace in the majority of rock pore systems, they become relevant to assess NMR measurements when the following three conditions are met: (a) microporosity regions are present within the grains; (b) micropores are well-connected (not cemented) to outer macroporous regions exhibiting low surface-to-volume ratios, thereby allowing fluid diffusion between both pore scales; and (c) rock surface relaxivity is low enough to prevent the decay of proton magnetization within the macroporosity before protons can enter the microporous regions. In addition, fluid diffusivity must be sufficiently large for diffusion coupling to be significant within the time scale of NMR measurements. This is usually the case only for water and light hydrocarbons. As a result, fluid magnetization will be exchanged between micro- and macropore regions, and no obvious relationship will exist between NMR transverse relaxation (T_2) distribution and pore-size distribution. Fig. 1 illustrates conditions (a) and (b), described previously, with an example of scanning electron microscope (SEM) images of a carbonate rock exhibiting diffusion coupling.

Varying TE is a common NMR data-acquisition technique used for in-situ reservoir fluid identification.¹⁻³ However, in the case

of NMR measurements performed in carbonate rocks, there are no published reports dealing with the impact of diffusion coupling on hydrocarbon typing and quantification using multiple- TE logging techniques. The objectives of this paper are twofold: (a) to develop a simulation algorithm capable of reproducing NMR measurements in complex pore geometries under a variety of experimental conditions and (in particular) under the influence of a constant magnetic field gradient, and (b) to provide simulation examples that will help assess the validity of fluid phase discrimination using multi- TE measurements in porous media exhibiting diffusion coupling.

In the past, numerical models of NMR decay were proposed based on periodic bimodal packs of spheres that accounted for surface relaxation effects. We have reproduced and extended Ramakrishnan *et al.*⁴'s Monte Carlo algorithm to account for the effect of an external constant magnetic field gradient of the type enforced by modern NMR tools, and in the presence of a nonwetting phase. The first part of this paper introduces the Monte Carlo simulation algorithm applicable to a bimodal pack of spheres in the presence of a constant magnetic field gradient. A subsequent section describes examples of numerical simulation that illustrate the versatility of the algorithm. Finally, we derive and interpret simulation examples intended to address specific issues of fluid discrimination using multiple interecho times in the presence of diffusion coupling. We address four specific cases of NMR T_2 distributions (two unimodal and two bimodal) by modeling several possible combinations of pore structure, diffusion coupling, and fluid distribution. These case studies were designed so that different pore configurations created identical NMR signals at low values of TE but exhibited differences at high values of TE .

Model for the Simulation of NMR Decay

The algorithm developed to numerically simulate NMR magnetization decay in carbonate rocks makes use of conditional Monte Carlo random walks. It is based on the algorithm described by Ramakrishnan *et al.*,⁴ further generalized to include microscopic diffusion effects in the presence of a constant magnetic field gradient.⁵ The assumption of a constant gradient across the zone probed by NMR tools is generally an accurate approximation in the presence of small contrasts of magnetic susceptibility (i.e., in the absence of paramagnetic materials). Porous media can include micro- and macroporous regions (grainstone model) to form a bimodal pore distribution, or they can exhibit a single pore size with solid grains (wackestone model).

Model Geometry. The simulation model was constructed with a 3D bimodal pack of spheres that replicates the complex pore-structure geometry encountered in some carbonate formations, namely microporous grainstones, where the solid micrograins are arranged into packs. As in the example shown in Fig. 1, the model adopted for the simulations accounts for a microporosity region immediately surrounding the grains, while different types of macroporosity (characterized by different sizes and degrees of cementation, depending on diagenesis) exist between the packs. The geometrical construction of the 3D model of a bimodal pack of spheres is based on the following assumptions:

- Bimodal pore-size distribution.
- Periodic geometry.
- Isotropic pores.

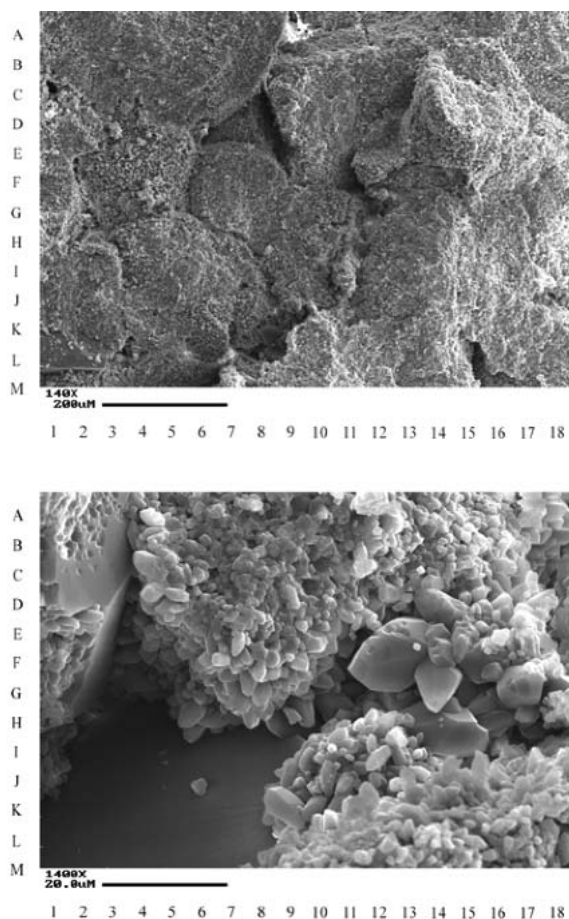


Fig. 1—SEM images of a carbonate sample at 140X (top panel) and 1400X (bottom panel) magnifications. A dual-porosity structure is clearly evidenced at these two scales: from a diagenetic viewpoint, the grains seen on the top panel were micritized into micrograins, as shown on the bottom panel. Therefore, the solid matrix is formed by micrograins agglomerated into packs or pseudo (microporous) macrograins.

- Spherical, compacted micrograins.
- Spherical, compacted grain packs.

At the micro- (or macro-) scale, each grain (or grain pack) of the model is defined as a sphere inscribed into a concentric cubic cell. If the sphere does not completely fill its cubic unit, then the complementary volume is considered filled with fluids. Furthermore, fixed blobs centered in the macropore space can be included in the model to represent partial saturations of the immiscible nonwetting oil phase, while the microporosity can only be filled with irreducible water. **Fig. 2** shows a 3D period of the assumed bimodal pore structure. Despite its simplicity, this model is able to reproduce important percolation properties that implicitly take into account restricted diffusion in both wetting and nonwetting fluids. Such properties include the following:

- The constrained relationship between pore-throat size and pore-body size creates a permeability percolation threshold; for porosities below 3%, all throats are closed to fluid passage.
 - The oil phase becomes spatially continuous only after the oil blobs reach the boundaries of the larger cubic cells.
- Diffusion behavior is best described as a very near-field phenomenon. Therefore, a model of periodic sphere packs within cubic grain packs is expected to reproduce diffusion phenomena within cubic grain packs similar to those taking place within disordered packs, as long as the distribution of pore size and throat size remain comparable in both cases.

Obviously, the periodic model adopted in this paper can only model one single macropore size and one single micropore size. This will result in NMR surface relaxation T_2 peaks sharper than those associated with a disordered sphere pack.

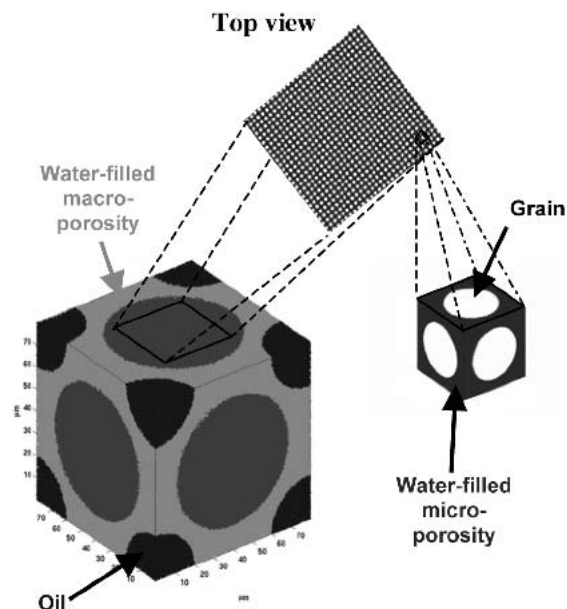


Fig. 2—Three-dimensional representation of the distribution of the fluids and rock matrix at the spatial resolution of both macro- and microcells. Gas blobs are centered in the macropores. Grain compaction is simulated by assigning sphere diameters larger than the concentric cube sizes.

Unimodal pore-size distributions can be constructed with the same basic model by allowing no microporosity. The NMR response of uncoupled bimodal pore structures (i.e., porous media where no diffusion coupling occurs) can be simulated by superimposing the response of each isolated pore mode.

Dual Random-Walk Strategy. At each iteration of the NMR numerical simulation algorithm, a fictitious proton is randomly chosen within the volume occupied by a fluid in the pore space. This proton is thereafter displaced to a new location determined by a continuous random walk operating according to two possible strategies. Depending on the proximity of the proton to the surface of material discontinuity (pore-wall or fluid-phase interface), either a conventional random-walk strategy or Zheng and Chiew's First-Passage-Time technique⁶ can be implemented to displace the proton to its next location.

The First-Passage-Time strategy was developed to allow macroscopic jumps in the bulk pore space and hence to speed up the otherwise computationally prohibitive random-walk process. It is based on the analytical solution of the unbounded diffusivity equation, which yields a stochastic relationship between the mean square displacement, $\langle R^2 \rangle$, during a given step, and its duration, Δt , with a maximum of probability close to the free space bulk diffusivity; that is,

$$D_o = \frac{\langle R^2 \rangle}{6\Delta t} \dots \dots \dots (1)$$

Fig. 3 illustrates the selection of the random-walk strategy according to the distance between the particles and the surrounding fluid boundaries. Grain surface and fluid interfaces constitute the wetting-phase boundaries, whereas only the interface between fluids constitutes a nonwetting phase boundary.

Model of Diffusive Relaxation. In general, the apparent transversal relaxation time T_2 of the decay of magnetization signal can be written as

$$\frac{1}{T_2} = \frac{1}{T_{2S}} + \frac{1}{T_{2B}} + \frac{1}{T_{2D}}, \dots \dots \dots (2)$$

where T_{2S} is the surface relaxation, T_{2B} is the bulk relaxation, and T_{2D} is the decay rate caused by diffusion in the presence of a

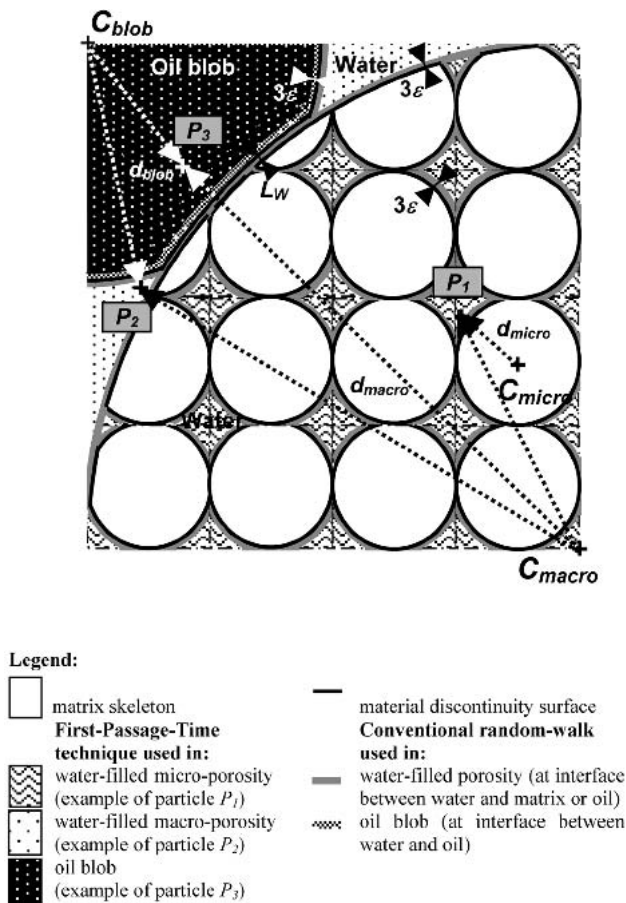


Fig. 3—Fraction of a cross section taken through a pack period. The graph illustrates the criteria used to select a specific random-walk technique according to the location of the particles at the onset of the walk step. A nominal distance of 3ϵ is taken as the safety thickness within which the First-Passage-Time technique reverses to a conventional random walk. A wetting layer of thickness L_w is also included to constrain the rock wettability at the surface of the microporous grain.

gradient field (diffusive relaxation). The Monte Carlo algorithm described by Ramakrishnan *et al.*⁴ takes into account the effect of surface relaxation, but it does not include the effect of bulk or diffusive relaxations. On the one hand, T_{2B} is governed by fluid properties, regardless of time or geometry, and therefore can be applied independently from the Monte Carlo simulation process. On the other hand, T_{2D} depends on both NMR experimental parameters (i.e., TE and gradient strength G) and the effective fluid diffusivity, D , as suggested by the macroscopic expression⁷

$$T_{2D} = \left(\frac{\gamma^2 G^2 TE^2 D}{12} \right)^{-1}, \dots \dots \dots (3)$$

where γ is the proton gyromagnetic ratio. When molecules diffuse through restricted geometries, the otherwise free diffusion is hindered; this causes the observed apparent diffusivity to become time- and geometry-dependent.⁸ The effective diffusivity D decays from the bulk (free-space) value D_o at a rate that depends on tortuosity, time, and pore surface-to-volume ratio. Within microporous regions in the micrometer range, for instance, the decay of D is instantaneous to reach an asymptotic value equal to the inverse of the tortuosity of the medium. By considering the diversity of pore sizes within carbonate rocks, it then becomes necessary to track the diffusivity effect through the random walk to model diffusive relaxation.

Because the magnetization signal acquired by the NMR tool is proportional to the cosine of the spin phases of the fluid protons,

we designed the algorithm so that it could track the spin dephasing caused by the constant magnetic field gradient. At time t , a proton spin phase is given by the integral

$$\phi(t) = \int_0^t \gamma \mathbf{G} \cdot \mathbf{r}(t') dt' = \gamma G \int_0^t z(t') dt', \dots \dots \dots (4)$$

where $\mathbf{r}$ is the position of the proton, and the gradient vector $\mathbf{G}$ is assumed to point in the z -direction. Because of the stochastic nature of the displacement z determined with the First-Passage-Time technique, the evaluation of the integral of Eq. 4 is not trivial. To circumvent this problem, we made use of the spin phase-shift expression derived by Gudgjarthsson and Patz,⁹ which is based on a stochastic formulation of the phase shift created by the gradient. This phase-tracking strategy was designed to honor the CPMG sequence with a sampling rate TE , as described next.

Simulation Algorithm. The following algorithmic sequence was applied for each proton and at each random-walk step, i , to account for the selection of a displacement strategy and of a CPMG sampling protocol:

1. Determine the distance d to the closest material interface.
2. If d is large, then the proton movement is inscribed in free space, and the First-Passage-Time technique is applied.^{4,6} Because of the constraint of the TE sequence, the duration Δt is such that it cannot extend beyond one half- TE period.
3. If d is small, then the previous algorithm does not converge. Therefore, it must be substituted by a conventional random-walk strategy with fixed infinitesimal displacement ϵ .
4. Regardless of d , the phase shift acquired during this step is given by⁹

$$\begin{aligned} \phi(t_{i+1}) = \phi(t_i) + \gamma G \left\{ \frac{[z(t_i) - z(0)] + [z(t_{i+1}) - z(0)]}{2} \right\} (t_{i+1} - t_i) \\ + \gamma G \sqrt{\frac{D_o(t_{i+1} - t_i)^3}{2}} \text{Normal}(\cdot), \dots \dots \dots (5) \end{aligned}$$

where t_i , t_{i+1} , and 0 are the initial instant of the step, the final instant of the step, and the time origin, respectively. In Eq. 5, $\text{Normal}(\cdot)$ is a Gaussian probability density function of zero mean and unity variance.

5. When $t_{i+1} = nTE$, where n is an integer, the phase cosine (i.e., the magnetization of the particle at time nTE) is recorded and added to the sum of signals of all the protons simulated at time nTE .

6. When $t_{i+1} = (n+1/2)TE$, where n is an integer, the phase is reversed to honor the radio-frequency 180° repolarizing CPMG magnetic pulses. This procedure makes the assumption of perfect and instantaneous pulses.

7. If contact is made with a surface of material discontinuity (either grain or interface between immiscible fluids), the proton magnetization decays with a probability equal to¹⁰

$$p = \frac{2\rho\epsilon}{3D_o}, \dots \dots \dots (6)$$

where ρ is the surface relaxivity of the interface. Depending on the wetting fluid and the rock, ρ typically ranges from 1 to 20 $\mu\text{m/s}$ for a rock/fluid interface. Fluid/fluid interfaces generally exhibit negligible surface relaxivity. If no magnetization decay occurs, the proton rebounds specularly at the surface of the interface.

Once a sufficient number of proton trajectories are simulated with the Monte Carlo algorithm, convergence to the average time solution is ensured, and a total magnetization signal is output. Generally, a few hundred to a thousand proton trajectories are needed for convergence within porous media, depending on the modeled pore-size distribution. The magnetization signal is subsequently inverted into a T_2 distribution by means of a curvature-smoothing regularization¹¹ generally used to invert noisy experi-

mental NMR magnetization decays. **Fig. 4** is a flow chart that graphically describes the details of the Monte Carlo algorithm.

Testing and Generalization of the Algorithm

Effect of Surface Relaxivity on Diffusion Pore Coupling in the Absence of a Magnetic Field Gradient. To establish a reference for nongradient experiments, we first conducted NMR simulations to illustrate the effect of surface relaxivity ρ on diffusion coupling. An interecho time, $TE = 1$ ms, was adopted in the presence of a null magnetic field gradient. **Fig. 5** shows good agreement between our simulation results and those published by Ramakrishnan *et al.*⁴ for similar geometrical and fluid properties. A limitation exists to the accuracy of our simulations for small TE sampling in

the case of large values of ρ , however. The principle of diffusion pore coupling is brought to light by considering the spatial molecular dispersion phenomenon illustrated in **Fig. 6**. Diffusion pore coupling increases when ρ decreases: the smaller the value of ρ , the higher the proton exchange, and therefore the higher the magnetization exchange between pores.

Effect of Dual Fluid Saturation on the NMR Response of Coupled Pore Modes. When immiscible oil blobs are included in the model, the amount of wetting water available for coupled magnetization is naturally reduced. Therefore, the less the water saturation (S_w), the less the transfer of magnetization between the two pore scales, thereby rendering diffusion coupling negligible. This phenomenon is illustrated in **Fig. 7**. Again, our simulation

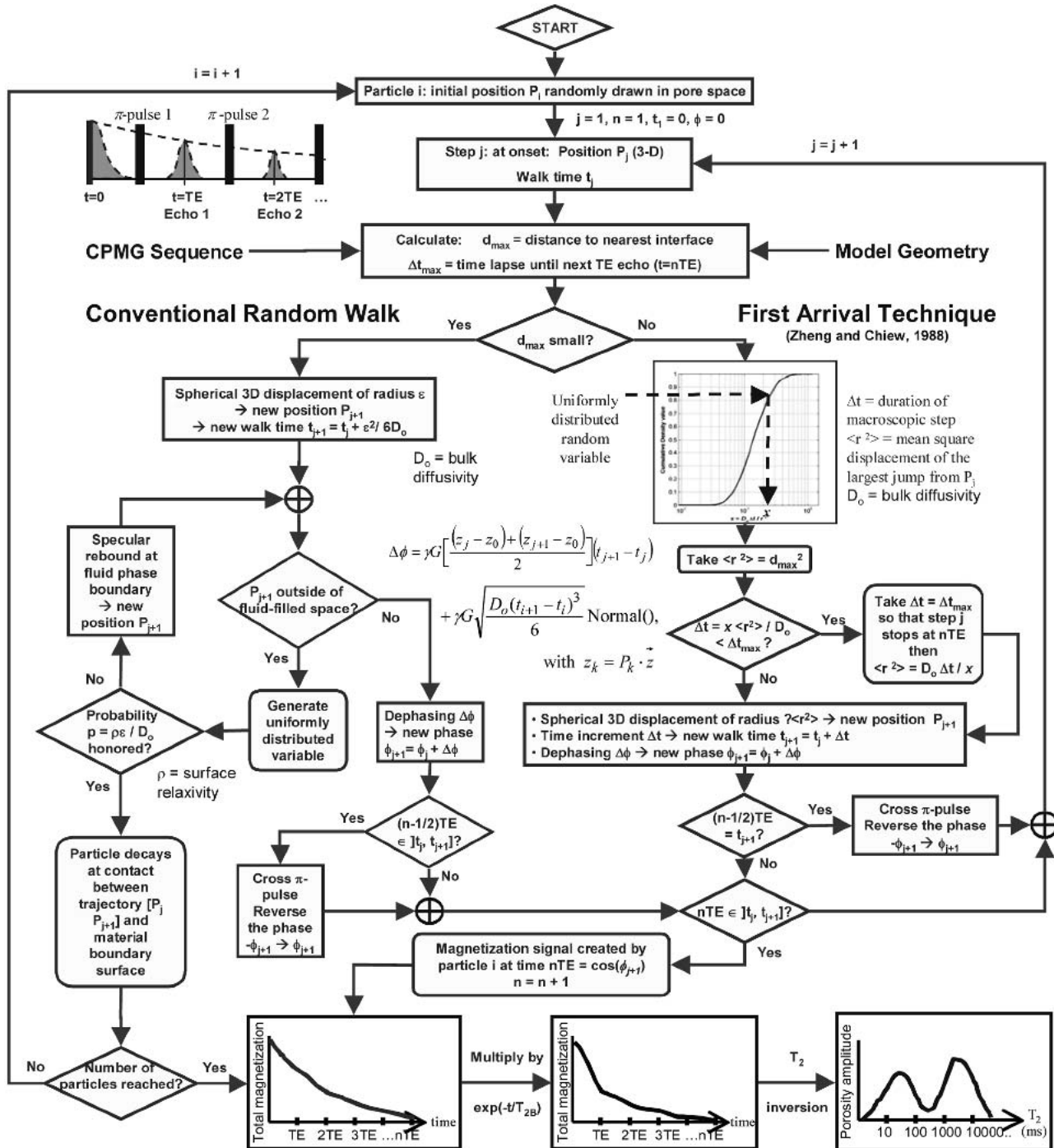


Fig. 4—Flow chart of the NMR simulation algorithm used for each fluid phase. In the descriptions, n is the echo index of the CPMG sampling, i is the particle index, j is the step index along particle i 's walk, P_j is the 3D position of particle i at the onset of step j , t_j is the walk time of particle i at the onset of step j , and ϕ_j is the spin phase of particle i at the onset of step j .

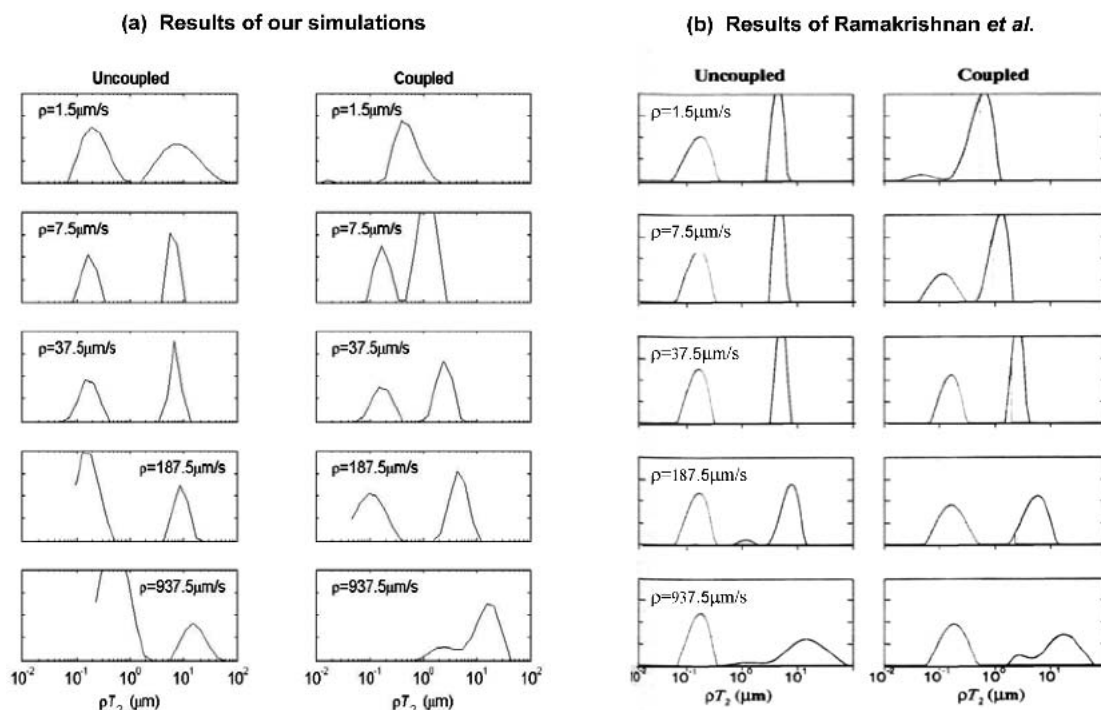


Fig. 5—Comparison of simulations of NMR data transformed into equivalent pore-size distributions: (a) simulations performed with our simulation program and (b) simulations reported by Ramakrishnan *et al.*,⁴ using a single water phase and different values of surface relaxivity ρ . Equivalent pore-size distributions were obtained by multiplying ρ and simulated T_2 distributions, and they exhibit the importance of ρ in NMR diffusion coupling. Although our algorithm is unable to detect low T_2 values in the theoretical case of high ρ values, the agreement between the two sets of simulations is remarkable.

results are in good agreement with those published by Ramakrishnan *et al.*⁴

Comparison With Analytical Relaxation in Free Diffusion. To test the consistency and accuracy of the spin-dephasing and phase-reversing strategies included in the NMR simulation algorithm,

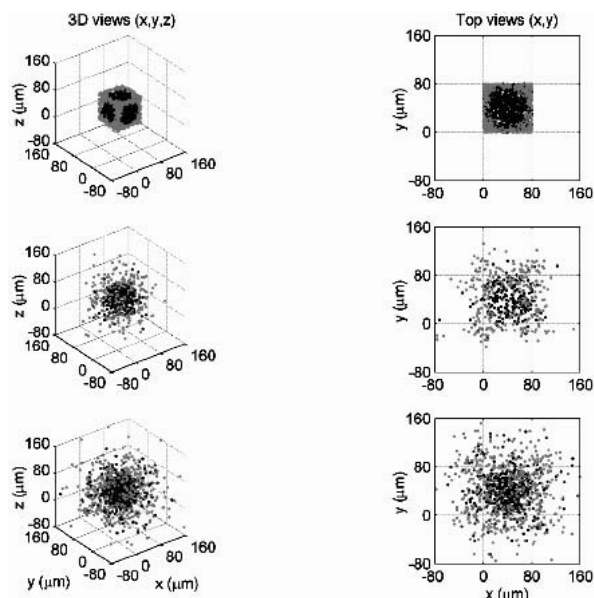


Fig. 6—Illustration of the spatial dispersion of fluid particles as a function of surface relaxivity, ρ . Initially, within a period of the dual pack of spheres, dark dots indicate locations of particles borne by the micropore space, whereas light dots indicate locations of particles borne by the macropore space. Final particle positions are either the locations of the particles at their decay or current locations of nondecayed particles after a 1-second lapse.

unbounded diffusion was simulated with grain diameters tending to zero for both 1-cp and 6-cp oils, at a temperature of 370 K, under a constant magnetic field gradient of strength 20 G/cm. Results from this simulation exercise are shown in Fig. 8. An

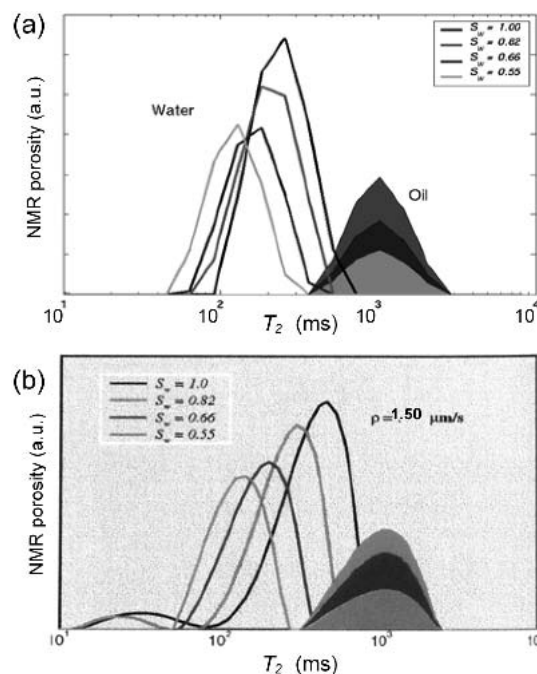


Fig. 7—Effect of nonwetting oil saturations on the NMR response of diffusively coupled systems. Simulation results of (a) our algorithm and (b) Ramakrishnan *et al.*'s⁴ results, obtained for similar coupled geometries. Gray-shaded peaks stand for the magnetization within the oil phase only, whereas gray lines limit the T_2 peaks of the magnetization signal caused by wetting water only.

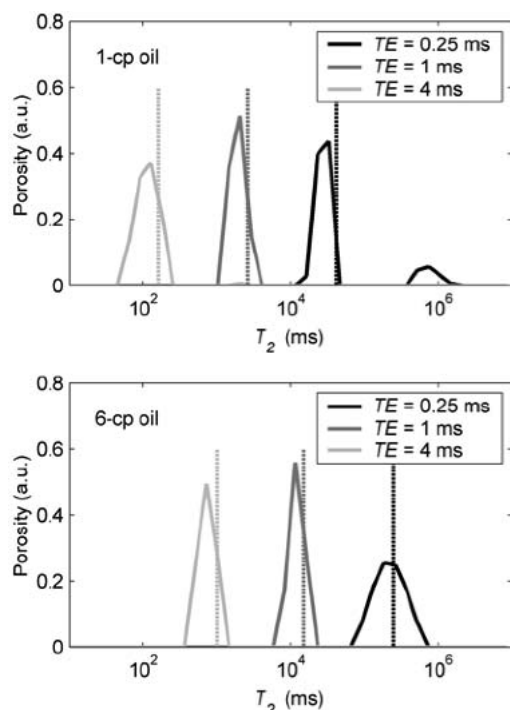


Fig. 8—Comparison of analytical T_{2D} relaxations (dash) and simulated T_2 distributions (solid) of 1-cp and 6-cp oil grades in unbounded diffusion. The simulation results are considered to account for no bulk relaxation [before multiplying the output decay by the function $\exp(-t/T_{2B})$]. T_{2D} values are calculated from Eq. 3 for $TE=0.25, 1$, and 4 ms [i.e., 160, 2,600, and 41,500 ms for light oil (1 cp), and 1,000, 15,000, and 248,000 ms for medium oil (6 cp), respectively].

excellent agreement is found between the macroscopic analytical expression given in Eq. 4 and the microscopic numerical results provided by our Monte Carlo simulation algorithm. Once bulk relaxation is applied to the results of the simulations, the match with analytical values [i.e., $1/(1/T_{2B} + 1/T_{2D})$] is even closer.

Example of Simulations of NMR Measurements Acquired in a Coupled Carbonate Rock. Under the assumption of diffusion coupling between existing pore modes, we reached very good agreements between experimental and simulation results. For instance, we were able to consistently calibrate the sphere-pack simulation model to geometrical pore core data.¹² Fig. 9 provides an example of such simulation results. The numerical simulations compare very well to experimental NMR measurements, including the case in which a magnetic field gradient is present. Based on the temperature dependence of the parameters D_o and T_{2B} used in the simulation algorithm, the model was used without an applied magnetic field gradient in conjunction with experimental data to assess the relative effects of diffusion coupling and temperature on NMR measurements and on the corresponding NMR-interpreted permeabilities.¹²

Application of the Simulation Algorithm to Fluid Discrimination With Multiple TE Measurements

NMR T_2 distributions acquired in carbonate formations often exhibit either a unimodal or a bimodal behavior. Other than the trivial case in which the saturating fluid is irreducible water, the interpretation of unimodal and bimodal distributions is not straightforward in the presence of non-negligible pore-coupling effects. Further, when both wetting (water) and nonwetting (oil) phases are present, the overall T_2 distribution is collectively influenced by all these mechanisms. Previous work on using multiple- TE logging measurements to identify oil saturation and viscosities¹⁻³ was based entirely on the assumption of uncoupled systems. Several NMR simulation examples were designed in oil/water systems to

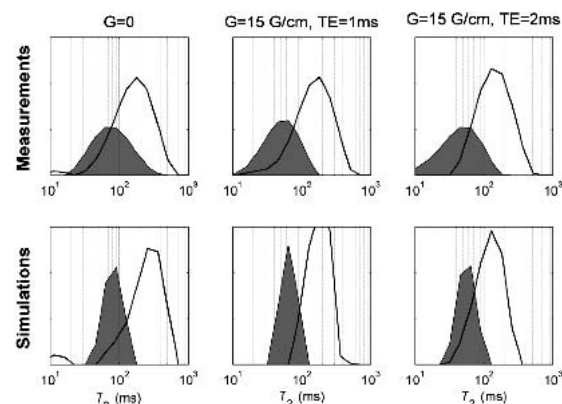


Fig. 9—Example of the comparison of measured and simulated T_2 distributions in the absence of a magnetic field gradient¹² ($G=0$), and then with ($G=15$ G/cm), for echo times of 1 and 2 ms. Gray-shaded (or unshaded) distributions correspond to measurements and simulations performed at irreducible water saturation (or 100% brine saturation). A very good agreement exists in terms of distribution spread for 100% brine saturation and T_2 peak value for both saturations, especially considering the limitations of the exclusively bimodal distribution of the model.

illustrate the capabilities and limitations of multiple- TE acquisition methods in the presence of complex pore structures and multiphase fluid distributions.

We consider four case studies for this simulation/interpretation component of the paper. The four cases are classified according to the T_2 distribution pattern simulated for standard interecho times of 0.25 ms, and they assume a magnetic field gradient strength of 15 G/cm. For each case, the standard NMR response is ambiguous and can be interpreted in very different ways using several hypotheses of distributions of pore sizes, structures, and saturating fluids. Simulation results are obtained for interecho times of 1 and 4 ms. According to the hypotheses formulated below, the models used in the simulations account for either a single pore size (for unimodal and uncoupled bimodal structures) or two coupled pore sizes (for coupled bimodal structures). The nominal pore sizes, as defined on the corresponding pore-size distributions of Figs. 10 through 13, were obtained by adjusting the geometrical dimensions of the unit cells of the periodic models. In all cases, the ratio between grain diameter and concentric cube side was kept constant and equal to 1.23, the surface relaxivity at the interface between fluids was neglected, and the grain surface relaxivity was held equal to $3 \mu\text{m/s}$, which corresponds to the usual surface relaxivities encountered in carbonates. Both light (1.5 cp) and medium (6 cp) oil grades in typical laboratory conditions were considered as nonwetting fluids, with bulk properties that are summarized in Table 1. In Figs. 10 through 13, T_2 distributions resulting from the simulation of the combined effect of bulk, surface, and diffusion relaxations are shown for each case. Gray-shaded T_2 distributions correspond to the contribution of oil only, while nonshaded distributions correspond to the total magnetization signal created by both oil and water. Because this combined signal is averaged on both fluids, the T_2 contribution of oil is smeared when it is included in the overall T_2 distribution within both fluids. This is why T_2 peaks corresponding only to oil look sharper and sometimes exhibit higher amplitudes than T_2 distributions of the total signal.

Case 1: Single T_2 Peak at High T_2 Values (1,000 ms) (Fig. 10). Hypothesis 1 (C1H1). Unimodal pore-size distribution filled only with water. The T_2 peak shifts to the left because of the increased role played by T_{2D} on the NMR time relaxation for $TE=4$ ms.

Hypothesis 2 (C1H2). Same as C1H1, but including blobs of light oil in the pores. At low TE values, the light-oil bulk relaxation equals the surface relaxation of the water filling the rest of the pore space. In comparison with C1H1, this oil grade decays slower than water with increasing values of TE . However, the difference in total signal is not noticeable.

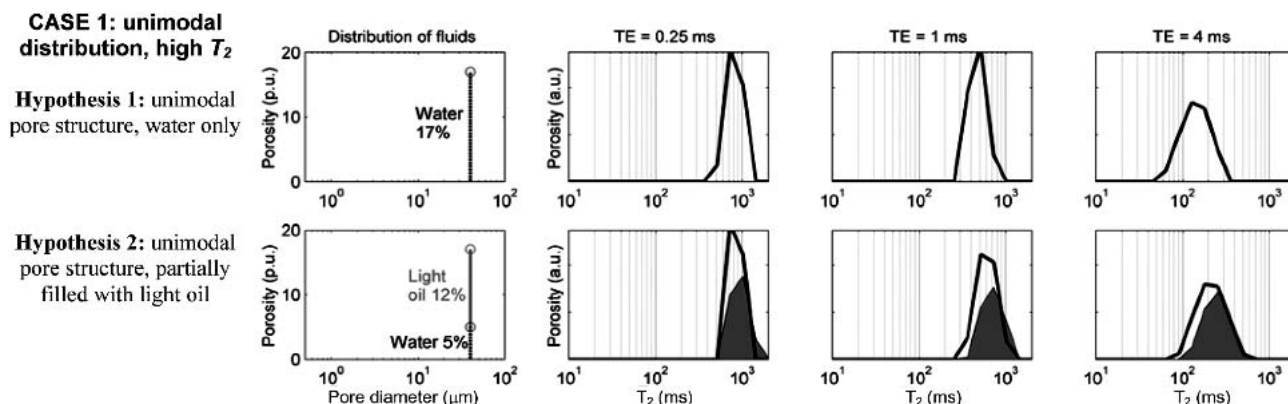


Fig. 10—Case 1 simulation results. Each row states a hypothesis, symbolizes it in terms of a fluid distribution within the pore system, and presents the results of the corresponding simulations for three values of interecho time. The first column of graphs summarizes the distribution of fluids in the pore space. Dashed (or plain) stems represent the porosity fractions of water (or oil) in the pore space characterized by the nominal pore sizes described in the horizontal axis of the plot. Fluid content is indicated in porosity units. The remaining columns describe the T_2 distributions resulting from the simulations with a 15 G/cm gradient, as well as TE values of 0.25, 1, and 4 ms. Gray-shaded T_2 distributions correspond to the magnetization within oil only, whereas unshaded T_2 distributions represent the magnetization within the whole pore system (i.e., including all fluid phases). Total magnetization can be understood as that measured with the NMR tool.

Case 2: Single T_2 Peak at Average T_2 Values (200 ms) (Fig. 11). **Hypothesis 1 (C2H1).** Unimodal pore-size distribution filled only with water. The nominal pore size is smaller than in C1H1.

Hypothesis 2 (C2H2). Comparable to C1H2, but with a medium-grade oil and a slightly smaller pore size than in C1H2. Oil bulk relaxation is assumed equal to water surface relaxation within the complementary pore space. The dominant oil saturation dictates the NMR behavior of the entire system. Because the bulk relaxation of the prominent oil saturation has a magnitude similar to its diffusive relaxation in bulk space (i.e., $D-D_o$), the increment in echo time only slightly modifies its T_2 peak value. We remark that the total black curve shown in the corresponding panel of Fig. 12 represents the joint response caused by both fluids and masks the NMR response of water peaking at $T_2 = 100$ ms (as in case C2H1).

Hypothesis 3 (C2H3). Bimodal, coupled pore structure with micro- and macroporosity regions, containing water only. This pore system behaves in a fashion similar to the unimodal C2H1

case described earlier, except for the case $TE = 4$ ms, in which the relaxation time corresponding to the T_2 peak is equal to half of that obtained for C2H1 (55 ms vs. 100 ms). This means that diffusively coupled pores manifest themselves as longer diffusion lengths, whereupon their magnetization signal decays faster than the equivalent signal because of uncoupled systems. In uncoupled systems, the macropores maintain the total T_2 response at values higher than that of micropores, whereas in coupled systems the contribution caused by micropores dominates the diffusive process, thereby leading to lower decay times.

Hypothesis 4 (C2H4). Bimodal, coupled pore structure, including medium-grade oil in the macroporous regions. The oil bulk relaxation is close to that of the coupled response described for C2H3. The presence of a medium oil maintains the relaxation value of the T_2 peak of the entire system higher than its value in C2H3 and, hence, equal to that of the unimodal pores of case C2H1. Because oil occupies part of the macropore space, less space is available for water, whereupon diffusion coupling be-

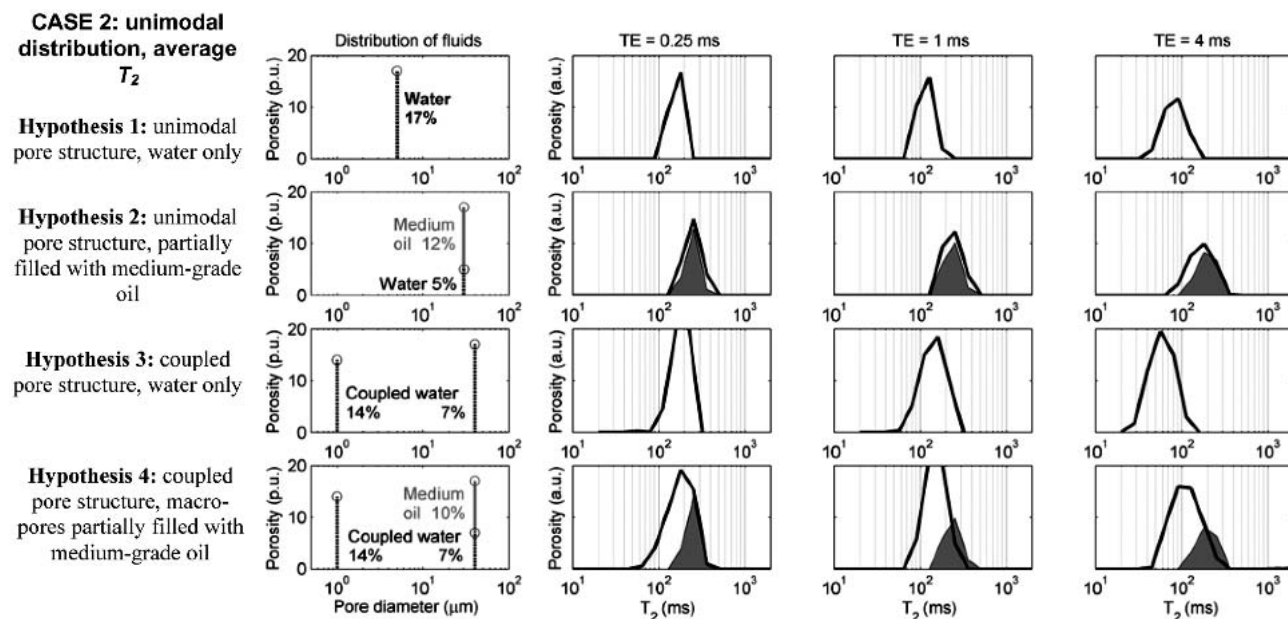


Fig. 11—Case 2 simulation results. Refer to the caption of Fig. 10 for details on the plotting conventions. Gray-shaded T_2 distributions correspond to the magnetization within oil only, whereas unfilled distributions represent the magnetization within the whole pore system (i.e., also including water).

CASE 3: bimodal distribution, low/high T_2

Hypothesis 1: uncoupled pore structure, water only

Hypothesis 2: uncoupled pore structure, macro-pores partially filled with medium oil

Hypothesis 3: uncoupled pore structure, macro-pores partially filled with light oil

Hypothesis 4: coupled or uncoupled pore structure, macro-pores totally filled with light oil

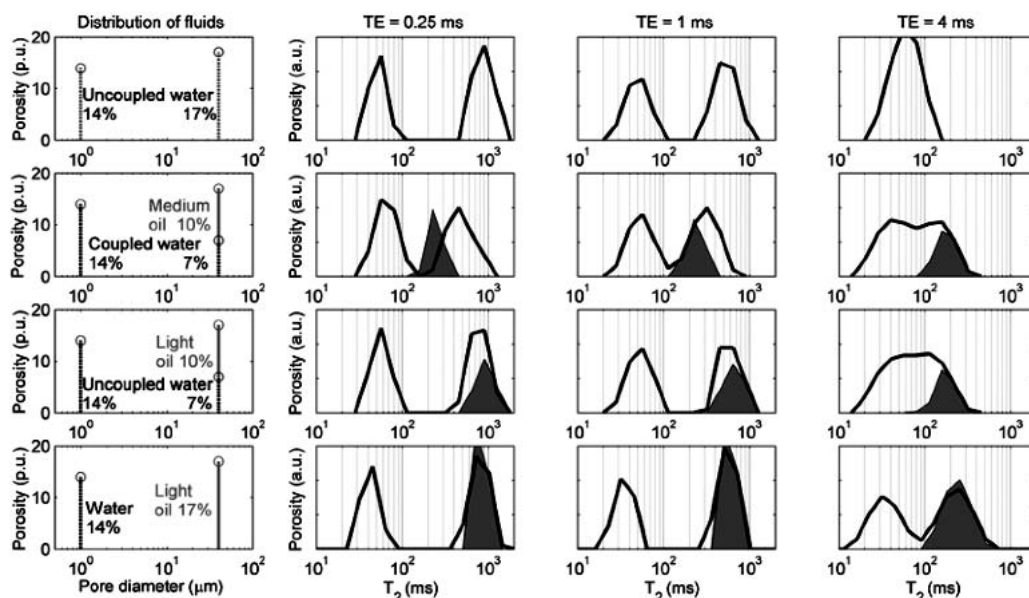


Fig. 12—Case 3 simulation results. Refer to the caption of Fig. 10 for details on the plotting conventions. Gray-shaded T_2 distributions correspond to the oil only, whereas unfilled distributions correspond to the whole pore system (also including water).

comes weak. On the one hand, the presence of residual water causes lower decay times because the proportion of microporous borne water is higher in the coupling mode. On the other hand, the oil phase exhibits a T_2 shift much lower than that of water when TE increases. Collectively, these two adverse effects make the T_2 response of the system slightly more spread than that of case C2H1.

Case 3: Double T_2 Peak at Low and High Values of T_2 (50 and 1,000 ms) (Fig. 12). As mentioned previously, the short T_2 peak can be caused solely by irreducible fluid. Hence, this configuration can only be representative of uncoupled pore systems or coupled

systems that remain effectively uncoupled by the complete oil saturation of the macropores. Therefore, this case is more typical of water/oil mixtures.

Hypothesis 1 (C3H1). Bimodal uncoupled pore-size distribution, filled with water only. The two distinct T_2 peaks originating from the pore-size segregation converge into a single peak when T_{2D} dominates the NMR response of the system.

Hypothesis 2 (C3H2). Bimodal uncoupled pore structure with partial saturation of medium oil. Because of the contrast in the relaxation times of the two fluids, for $TE = 0.25$ ms, the T_2 value of its macroporosity peak shifts from 1,000 ms for full water

CASE 4: bimodal distribution, average/high T_2

Hypothesis 1: uncoupled pore structure, water only

Hypothesis 2: coupled or uncoupled pore structure, macro-pores completely filled with light oil

Hypothesis 3: unimodal pore structure, partially filled with light oil

Hypothesis 4: coupled pore structure, macro-pores partially filled with light oil

Hypothesis 5: uncoupled pore structure, macro-pores partially filled with light oil

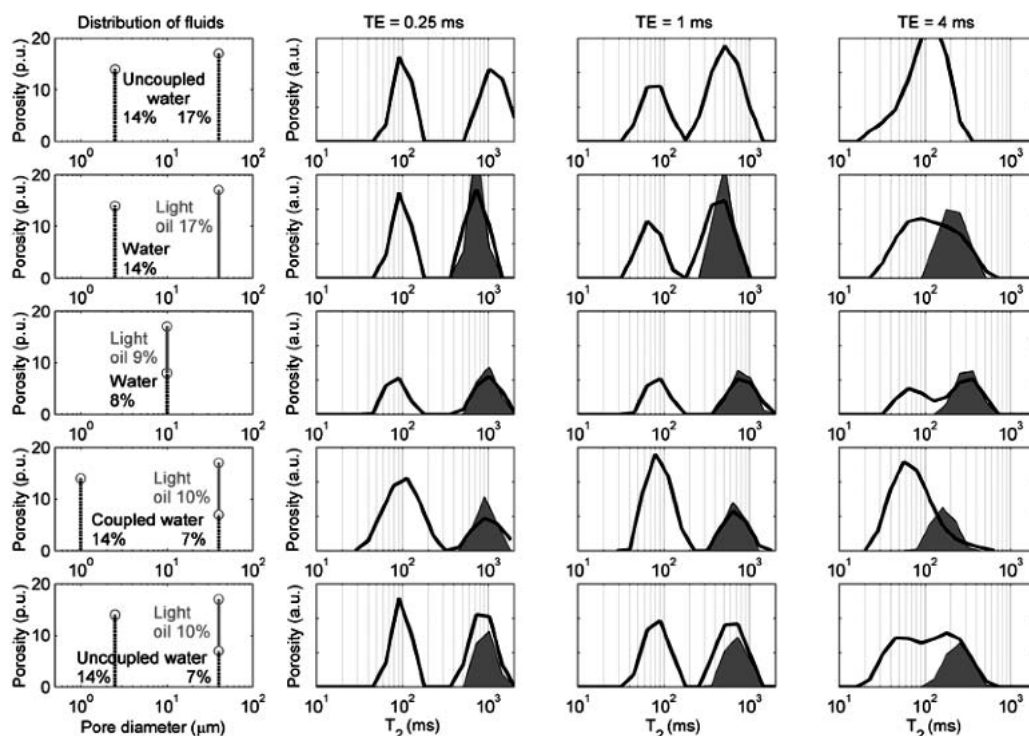


Fig. 13—Case 4 simulation results. Refer to the caption of Fig. 10 for additional details on the plotting conventions. Gray-shaded T_2 distributions correspond to the oil only, whereas unfilled distributions correspond to the whole pore system (also including water).

TABLE 1—SUMMARY OF BULK FLUID PROPERTIES USED FOR THE SIMULATION OF NMR FLUID SATURATION EFFECTS

Fluid	$\mu(\text{cp})$ at 370 K	$D_o (\mu\text{m}^2/\text{ms})$	$T_{2B} (\text{ms})$
Brine (water)	1	2.5	3,700
Light oil	1.5	1	1,000
Medium oil	6	0.27	250

saturation (see C3H1) to 500 ms for partial oil saturation. When TE increases, this shift is reinforced and splits the magnetization between two modes for $TE = 4$ ms.

Hypothesis 3 (C3H3). Same configuration as in C3H2, but with light oil. The fact that light-oil bulk relaxation is closer to 1,000 ms than medium-oil bulk relaxation causes the NMR response at low echo time to be the same as that for full water saturation (C3H1). However, a large T_2 spread appears at the longest echo times where, on the contrary, water alone causes a sharp, selective T_2 decay. Despite different values of T_{2B} between this light oil and the medium oil used in case C3H2, at $TE = 4$ ms the effect of the applied magnetic field cancels any possibility of discrimination based on total relaxation time. Efficient discrimination of both oil types is not likely with multiple echo times in this geometrical pore configuration. Echo times longer than 4 ms would be required to do so, but these would entail excessively large levels of measurement noise.

Hypothesis 4 (C3H4). Bimodal coupled or uncoupled pore structure, with complete light-oil saturation of the macropores. This case is mainly characterized by a complete segregation of the oil and water T_2 peaks at large echo times. Given the large contrast in size between micro- and macropores (a factor of 40), such a situation provides obvious evidence of the complete saturation of the macropores with light oil (see also C4H2). Case C3H4 represents a good-quality oil reservoir with no free water, in which case diffusive coupling has a negligible impact on NMR interpretation.

Case 4: Double T_2 Peak at Average and High Values of T_2 (100 and 1,000 ms) (Fig. 13). **Hypothesis 1 (C4H1).** Bimodal uncoupled system, same as C3H1 but with larger micropores (which become mesopores).

Hypothesis 2 (C4H2). Same configuration as C3H4, with light oil and mesopores instead of micropores. The ratio of macro- to mesopore size is smaller (i.e., 15) than the ratio of macro- to micropore size in case C3H4 (i.e., 40). This difference is enough to explain the fact that in case C3H4, two distinct peaks appear for $TE = 4$ ms, whereas in case C4H2, water magnetization exhibits decay times close to those of oil magnetization. As a consequence, the total system response exhibits a single, largely spread T_2 peak. It can be concluded that the reservoir quality interpreted in case C3H4 (two distinct T_2 peaks at long TE) is valid only in the presence of microporosity.

Hypothesis 3 (C4H3). Unimodal pore structure partially filled with light oil. The main objective of this simulation case is to explore a possible difference with case C4H2 where, despite the same effective pore size left for water saturation (same peak relaxation at $TE = 0.25$ ms), the actual pore size and structure are completely different. It is observed that the unimodal pore structure exhibits a response at long echo times. Consequently, a more pronounced difference arises between the oil peak and the water peak. However, as shown by case C3H4, the pattern is not unique.

Hypothesis 4 (C4H4). Bimodal, diffusively coupled pore structure partially saturated with light oil. As discussed earlier, a 100-ms T_2 peak can be the result of diffusion coupling between micro- and macropores. In the presence of both irreducible water and partial light-oil saturations, when TE increases, this coupled system exhibits an important transfer of magnetization toward short relaxation times. Transfer of magnetization is such that the presence of oil is only suggested by a tail toward long T_2 of the total response at $TE = 4$ ms.

Hypothesis 5 (C4H5). This case is constructed with the same uncoupled pore configuration as in C4H1, except for the partial saturation of the macroporous region with light oil. By direct comparison with the simulation results obtained for case C4H4, we remark that, as with cases C3H2 and C3H3, the coupled case C4H4 is not more amenable to fluid-phase discrimination than the uncoupled case C4H5. However, this fundamental difference, based on the transfer of the macroporous borne magnetization to shorter values of T_2 , suggests a practical way to discriminate between the coupled or uncoupled nature of the pore structure.

Conclusions

A novel numerical algorithm was developed to simulate NMR measurements acquired in the presence of a constant magnetic field gradient. The geometrical and petrophysical model of rock formations included dual fluid saturations and dual-porosity structures exhibiting diffusion coupling.

The algorithm was used to assess the sensitivity of NMR acquisition strategies based on measurements with multiple echo times. These measurements are generally used in the detection and quantification of partial hydrocarbon saturation in rock formations. We focused our simulation and interpretation work on carbonate-like rocks in which sufficient contrast existed between the molecular diffusion coefficients of the saturating fluids. It was found that long TE values could help to unambiguously characterize pore systems in which light oil fully saturated macropores and brine fully saturated micropores. However, as soon as diffusion coupling was present and multiple phases coexisted in the macropore space, multiple- TE NMR interpretation became ambiguous. The presence of diffusive coupling enhanced the contrast between water and oil signals. It was found, however, that information on the hydrocarbon grade was required to reach reliable conclusions. Overall, this exercise showed the value of versatile numerical simulation algorithms like the one described in this paper in the interpretation of petrophysical properties from NMR measurements.

Nomenclature

- D = effective diffusivity, m^2/s
- D_o = bulk diffusivity, m^2/s
- G = strength of external magnetic field gradient applied by NMR tool, G/cm
- p = probability of proton magnetization decay at fluid surface boundary, dimensionless
- R = macroscopic displacement of a step of First-Passage-Time technique, μm
- S/V = surface-to-volume ratio of pore, μm^{-1}
- S_w = water saturation, dimensionless
- t = time, ms
- TE = interecho time of the NMR tool, ms
- T_2 = total transverse relaxation, ms
- T_{2B} = bulk transverse relaxation, ms
- T_{2D} = diffusive transverse relaxation, ms
- T_{2S} = surface transverse relaxation, ms
- Δt = duration associated with a First-Passage-Time technique step, ms
- ε = unitary displacement of a step of conventional random-walk technique applied near the pore wall, nm
- γ = proton gyromagnetic ratio = $2\pi \cdot 4258 \text{ rad}/(\text{G}\cdot\text{s})$
- μ = fluid viscosity, cp
- ρ = surface relaxivity, $\mu\text{m}/\text{s}$
- ϕ = proton spin phase, rad

Acknowledgments

We express our gratitude to Baker Atlas for partially funding this work through an internship position offered to Emmanuel Toumelin during summer 2001. We would also like to thank Gigi Zhang and Carl Edwards for their useful discussions. Partial support is acknowledged from the Center of Excellence in Formation Evaluation of the U. of Texas at Austin and from the American Chemical

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SI Metric Conversion Factors

$\text{\AA} \times 1.0^*$	E-01 = nm
cp $\times 1.0^*$	E-03 = Pa·s
$\text{ft}^2 \times 9.290\,304^*$	E-02 = m ²
$^{\circ}\text{F} \quad (^{\circ}\text{F} + 459.67)/1.8$	= K
in. $\times 2.54^*$	E+00 = cm

*Conversion factor is exact.

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